

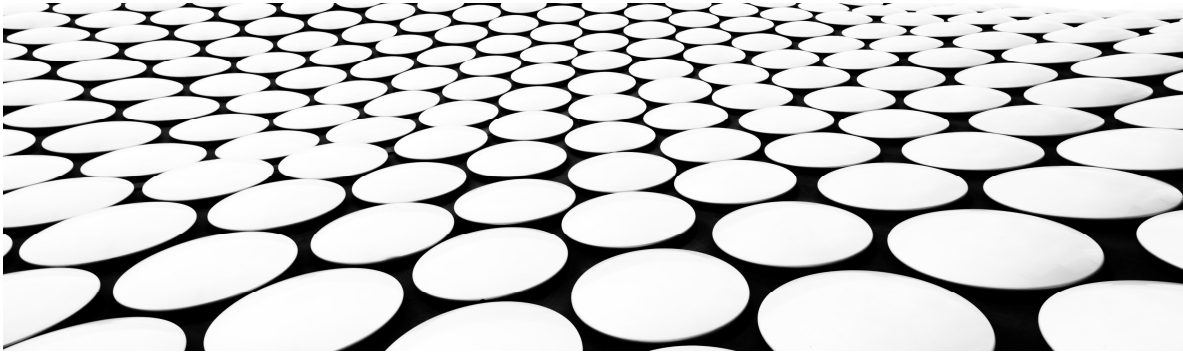


ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
KHOA ĐIỆN TỬ - VIỄN THÔNG

THIẾT KẾ VI MẠCH ĐIỆN TỬ

CHAPTER 5: LOGIC CMOS CIRCUITS

LÊ ĐỨC HÙNG



CONTENT

- CMOS Inverter
- CMOS NAND gates
- CMOS NOR gates
- CMOS Complex Gates

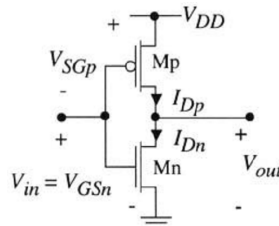
CMOS INVERTER

Basic Circuit and DC Operation

- CMOS Inverter (or NOT) is a basic digital circuit
- It consists of 2 opposite polarity MOSFET: NMOS and PMOS with their gates connected together at the input, the applied voltage is denoted by V_{in} . The inverter output voltage V_{out} is taken from the common drain terminals.
- An nFET-pFET group with a common gate is called a complementary pair, which gives us the "C" in "CMOS." The complementary pair forms the basis for CMOS logic circuits.
- The inverter circuit operation can be understood by examining the relationship between and the gate-source voltages of the FETs.

$$V_{GSn} = V_{in}$$

$$V_{SGp} = V_{DD} - V_{in}$$



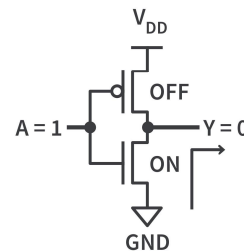
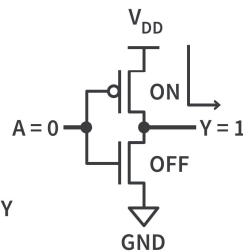
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CMOS INVERTER

Basic Circuit and DC Operation

- V_{IL} = input low voltage
- V_{IH} = input high voltage
- V_{OL} = output low voltage
- V_{OH} = output high voltage
- If $V_{in} = V_{DD}$, the p-channel MOSFET Mp is in cut-off while the n-channel MOSFET Mn is conducting in the non-saturated mode. Mn then provides a current path to ground, resulting in an output voltage
 - ✓ $V_{GSn} = V_{DD}$
 - ✓ $V_{SGp} = 0$
 - ✓ $\min(V_{out}) = V_{OL} = 0$
 - ✓ $V_{in} = V_{DD} \rightarrow V_{out} = 0V$
- If $V_{in} = 0$, the Mn is in cut-off while Mp is conducting in the non-saturated mode. Mp then provides a conductive path to power supply and gives
 - ✓ $V_{GSn} = 0$
 - ✓ $V_{SGp} = V_{DD}$
 - ✓ $\max(V_{out}) = V_{OH} = V_{DD}$
 - ✓ $V_{in} = 0V \rightarrow V_{out} = V_{DD}$

A	Y
0	1
1	0



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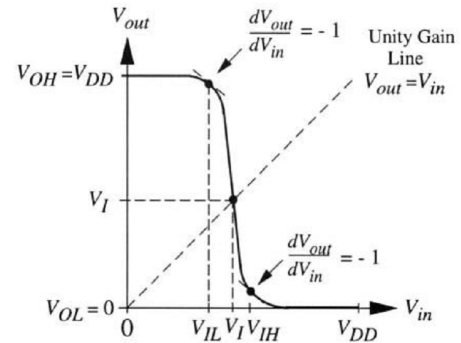
CMOS INVERTER

Voltage Transfer Curve (VTC)

- Plot of V_{out} as a function of V_{in} . It shows DC input-output characteristics. Vary V_{in} from 0 to V_{DD} .
- All of the important DC circuit characteristics can be extracted from the VTC.
- The VTC itself gives a set of critical voltages to work with in creating the defined ranges. The important values are the output voltages V_{OL} and V_{OH} , and the input voltages V_{IL} and V_{IH} , and the inverter threshold voltage V_I .

$$\beta_n = K'_N \left(\frac{W}{L} \right)_n$$

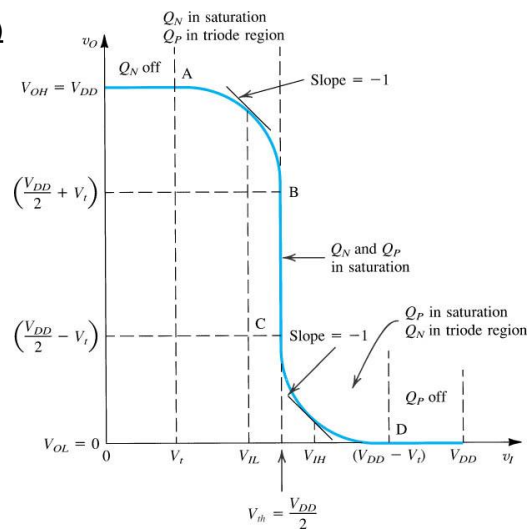
$$\beta_p = K'_P \left(\frac{W}{L} \right)_p$$



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CMOS INVERTER

Voltage Transfer Curve (VTC)

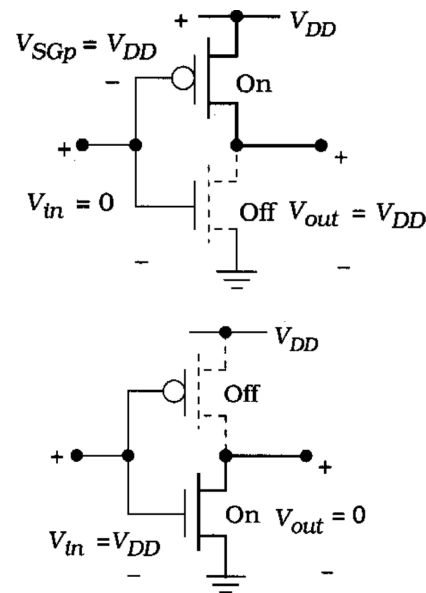


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CMOS INVERTER

Basic Circuit and DC Operation

- Output High Voltage, V_{OH}
 - maximum output voltage
 - occurs when input is low ($V_{in} = 0V$)
 - pMOS is ON, nMOS is OFF
 - pMOS pulls V_{out} to V_{DD}
 - $V_{OH} = V_{DD}$
- Output Low Voltage, V_{OL}
 - minimum output voltage
 - occurs when input is high ($V_{in} = V_{DD}$)
 - pMOS is OFF, nMOS is ON
 - nMOS pulls V_{out} to Ground
 - $V_{OL} = 0V$
- Logic Swing
 - Max swing of output signal
 - $V_L = V_{OH} - V_{OL}$
 - $V_L = V_{DD}$



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CMOS INVERTER

DC Characteristics

- **Output high voltage (V_{OH})** = the largest value of V_{out}
- Applying an input voltage $V_{in} < V_{Tn}$ which insures that the nFET is in cutoff while the pFET is biased into the active region. Ideally, the simplified MOSFET equations give $I_{dp} = 0$ which implies that the source-drain voltage $V_{SDp} = 0$.
 - $V_{out} = V_{DD} - V_{SDp} = V_{DD} = V_{OH}$
- **Output low voltage (V_{OL})** = the smallest value of V_{out}
- Setting the input voltage to a value $V_{in} = V_{DD}$ places Mp in cutoff and Mn is in active region. Since Mn is biased active but has $I_{Dn} = 0$, the drain-source voltage across the nMOSFET is $V_{DSn} = 0$.
 - $V_{out} = V_{DSn} = 0 = V_{OL}$

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CMOS INVERTER

- **Input low voltage (V_{IL})** = the largest value of V_{in} that can be interpreted as a logic 0 input
- If $V_{in} < V_{IL}$, then the output voltage V_{out} is either at V_{DD} or close to it, indicating that the output can be interpreted as a logic 1. If $V_{in} > V_{IL}$, the circuit moves into the transition region.

$$\left. \begin{aligned} \frac{\beta_n}{2}(V_{in} - V_{Tn})^2 &= \frac{\beta_p}{2}[2(V_{DD} - V_{in} - |V_{Tp}|)(V_{DD} - V_{out}) - (V_{DD} - V_{out})^2] \\ V_{in} \left(1 + \frac{\beta_n}{\beta_p}\right) &= 2V_{out} - V_{DD} - |V_{Tp}| + \frac{\beta_n}{\beta_p}V_{Tn} \end{aligned} \right\} \text{Solve for } V_{in} = V_{IL}$$

- **Input high voltage (V_{IH})** = the smallest value of V_{in} that can be interpreted as a logic 1 input
- If $V_{in} > V_{IH}$, the output voltage that is either 0 or close to it.
- Mn is non-saturated while Mp is saturated so that equating currents gives

$$\left. \begin{aligned} \frac{\beta_n}{2}[2(V_{in} - V_{Tn})V_{out} - V_{out}^2] &= \frac{\beta_p}{2}(V_{DD} - V_{in} - |V_{Tp}|)^2 \\ V_{in} \left(1 + \frac{\beta_p}{\beta_n}\right) &= 2V_{out} + V_{Tn} + \frac{\beta_p}{\beta_n}(V_{DD} - |V_{Tp}|) \end{aligned} \right\} \text{Solve for } V_{in} = V_{IH}$$

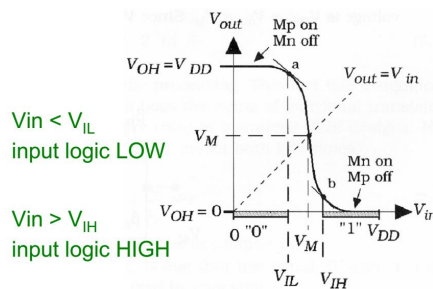
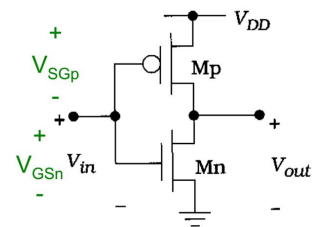
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CMOS INVERTER

Basic Circuit and DC Operation

- Gate Voltage, $f(V_{in})$
 - $V_{GSn} = V_{in}$, $V_{SGp} = V_{DD} - V_{in}$
- Drain Voltage, $f(V_{out})$
 - $V_{DSn} = V_{out}$, $V_{SDp} = V_{DD} - V_{out}$
- Transition Region (between V_{OH} and V_{OL})
 - V_{in} low
 - $V_{in} < V_{in}$
 - Mn in Cutoff, OFF
 - Mp in Triode, V_{out} pulled to VDD
 - $V_{in} > V_{in} > \sim V_{out}$
 - Mn in Saturation, strong current
 - Mp in Triode, V_{SG} & current reducing
 - V_{out} decreases via current through Mn

- $V_{in} = V_{out}$ (mid point) $\approx \frac{1}{2}V_{DD}$
 - Mn and Mp both in Saturation
 - Maximum current at $V_{in} = V_{out}$
- V_{in} high
 - $V_{in} > \sim V_{out}$, $V_{in} < V_{DD} - |V_{Tp}|$
 - Mn in Triode, Mp in Saturation
 - $V_{in} > V_{DD} - |V_{Tp}|$
 - Mn in Triode, Mp in Cutoff

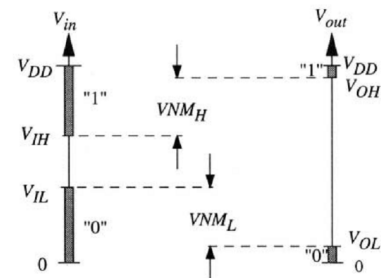
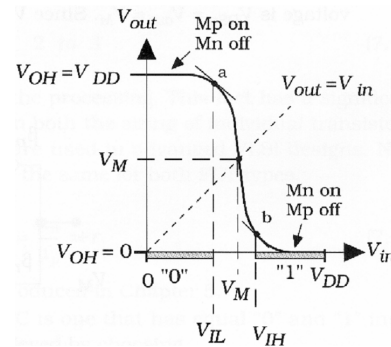


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CMOS INVERTER

Basic Circuit and DC operation

- Input Low Voltage, V_{IL}
 - V_{in} such that $V_{in} < V_{IL} = \text{logic 0}$
 - point 'a' on the plot
 - where slope, $\frac{\partial V_{in}}{\partial V_{out}} = -1$
- Input High Voltage, V_{IH}
 - V_{in} such that $V_{in} > V_{IH} = \text{logic 1}$
 - point 'b' on the plot
 - where slope = -1
- Voltage Noise Margins (VNM)
 - measure of how stable inputs are with respect to signal interference
 - $VNM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH} (V_{OH} = V_{DD})$
 - $VNM_L = V_{IL} - V_{OL} = V_{IL} (V_{OL} = 0)$
 - desire large VNM_H and VNM_L for best noise immunity



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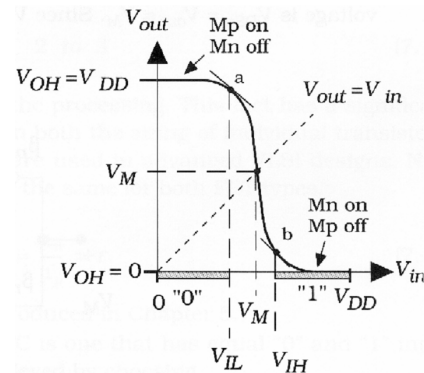
CMOS INVERTER

Switching Threshold

- Switching threshold = point on VTC where $V_{out} = V_{in}$
 - also called midpoint voltage, V_M
 - here, $V_{in} = V_{out} = V_M$
- Calculating V_M
 - at V_M , both nMOS and pMOS in Saturation
 - in an inverter, $I_{Dn} = I_{Dp}$
 - solve equation for V_M

$$I_{Dn} = \frac{\mu_n C_{ox} W}{2L} (V_{GSn} - V_{tn})^2 = \frac{\beta_n}{2} (V_{GSn} - V_{tn})^2 = \frac{\beta_p}{2} (V_{GSp} - |V_{tp}|)^2 = I_{Dp}$$

$$I_{Dp} = \frac{\beta_p}{2} (V_{GSp} - |V_{tp}|)^2$$



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CMOS INVERTER

Switching Threshold

- Calculating V_M

- $I_{Dn} = I_{Dp}$, $V_{in} = V_{out} = V_M = V_{GSn} = V_{DD} - V_{SGp} = V_{SDp}$
- solve equation for V_M

$$I_{Dn} = \frac{\mu_n C_{ox} W}{2L} (V_{GSn} - V_{tn})^2 = \frac{\beta_n}{2} (V_{GSn} - V_{tn})^2 = \frac{\beta_p}{2} (V_{SGp} - |V_{tp}|)^2 = I_{Dp}$$

- express in terms of V_M

$$\frac{\beta_n}{2} (V_M - V_{tn})^2 = \frac{\beta_p}{2} (V_{DD} - V_M - |V_{tp}|)^2 \rightarrow \sqrt{\frac{\beta_n}{\beta_p}} (V_M - V_{tn}) = V_{DD} - V_M - |V_{tp}|$$

- solve for V_M

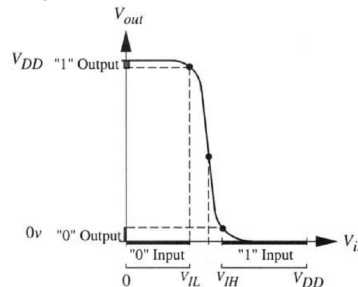
$$V_M = \frac{V_{DD} - |V_{tp}| + V_{tn} \sqrt{\frac{\beta_n}{\beta_p}}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

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CMOS INVERTER

Switching Threshold

- If $0 \leq V_{in} \leq V_{IL}$ ("0" input): $V_{out} = V_{DD}$ ("1" output)
- If $V_{IH} \leq V_{in} \leq V_{DD}$ ("1" input): $V_{out} = 0$ ("0" output)
- The inverter threshold voltage V_I : $V_{IL} \leq V_I \leq V_{IH}$. We may interpret this as the midpoint voltage such that
 - If $V_{in} < V_M \rightarrow$ Input is probably a logic 0
 - If $V_{in} > V_M \rightarrow$ Input is probably a logic 1



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Transistor size on VTC

- Recall

$$\beta_n = K'_N \left(\frac{W}{L}\right)_n \quad \frac{\beta_n}{\beta_p} = \frac{K'_N \left(\frac{W}{L}\right)_n}{K'_P \left(\frac{W}{L}\right)_p}$$

- If nMOS and pMOS are same size

$$(W/L)_n = (W/L)_p, C_{oxn} = C_{oxp} \rightarrow \frac{\beta_n}{\beta_p} = \frac{K'_N \left(\frac{W}{L}\right)_n}{K'_P \left(\frac{W}{L}\right)_p} = \frac{\mu_n C_{oxn} \left(\frac{W}{L}\right)_n}{\mu_p C_{oxp} \left(\frac{W}{L}\right)_p} = \frac{\mu_n}{\mu_p} \cong 2 \text{ or } 3$$

- If

$$\frac{\mu_n}{\mu_p} = \frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p} \rightarrow \frac{\beta_n}{\beta_p} = 1$$

since L normally min. size for all transistor,
can get betas equal by making W_p larger than W_n

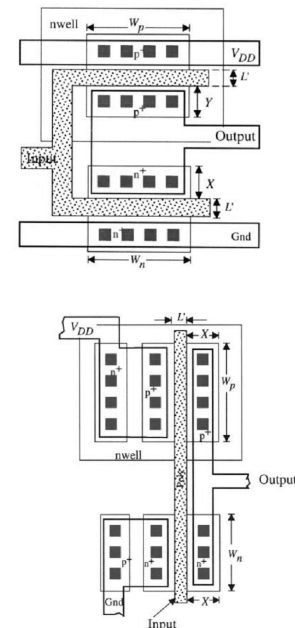
- Effect on switching threshold
 - If $\beta_n \approx \beta_p$ and $V_{tn} = |V_{tp}|$, $V_M = V_{DD}/2$, exactly in the middle.
- Effect on noise margin
 - If $\beta_n \approx \beta_p$, V_{IH} and V_{IL} both close to V_M and noise margin is good.

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CMOS INVERTER

Layout of Inverter

- In this example, the FETs have been chosen to have equal channel widths with $W_n = W_p$, so that $(W/L)_n = (W/L)_p$. The dimensions of the drain sections are shown as $(W_p \times Y)$ for the pFET region, and $(W_n \times X)$ for the nFET region. These values do not affect the DC operation, but enter into the calculation of the switching transients when finding the parasitic capacitance contributions that affect the circuit.
- This example employs different aspect ratios for the two transistors, with $W_p > W_n$. This is sometimes done to compensate for $k'_p < k'_n$. One important point to note from the DC analysis is that the critical voltages in the VTC depend only upon the ratio.

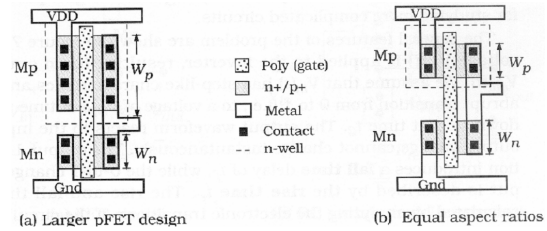


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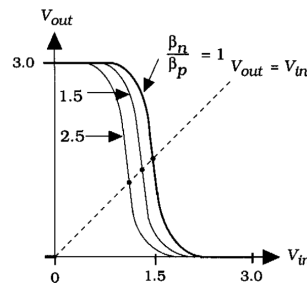
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Transistor size on VTC

- Can get betas equal by making W_p larger than W_n



- Transition pushed to left as beta ratio increases



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CMOS INVERTER

Example

- Let assume an inverter designed with $(W/L)_n = 8$ and $(W/L)_p = 12$, which is fabricated in a process where $k'_n = 100\mu\text{A/V}^2$, $V_{T0n} = 0.75\text{V}$, $k'_p = 40\mu\text{A/V}^2$, $V_{T0p} = -0.8\text{V}$
- Find the aspect ratios: $\beta_n = 800\mu\text{A/V}^2$, $\beta_p = 480\mu\text{A/V}^2 \rightarrow \beta_n/\beta_p = 1.67 > 1$
- Assume a power supply: $V_{DD} = 3.3\text{V} \rightarrow V_{OH} = V_{DD} = 3.3\text{V}$, $V_{OL} = 0\text{V}$

$$1.67(V_{IL} - 0.75)^2 = 2(2.5 - V_{IL})(3.3 - V_{out}) - (3.3 - V_{out})^2$$

$$V_{IL}(1 + 1.67) = 2V_{out} + (1.67)(0.75) - 2.5$$

- Solve for: $V_{IL} \approx 1.26\text{V}$ at the output voltage $V_{out} \approx 3.11\text{V}$

$$\left(\frac{1}{1.67}\right)(2.5 - V_{IH})^2 = 2(V_{IH} - 0.75)V_{out} - V_{out}^2$$

$$V_{IH}\left(1 + \frac{1}{1.67}\right) = 2V_{out} + \frac{1}{1.67}(2.5) + 0.75$$

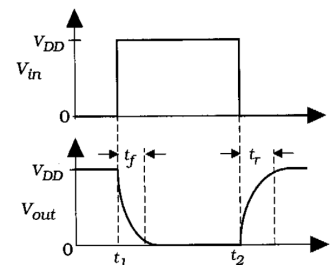
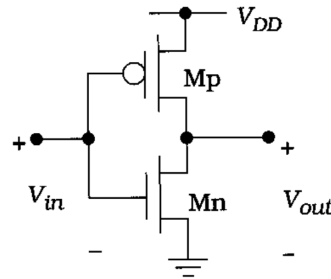
- Solve for: $V_{IH} \approx 1.7\text{V}$ at $V_{out} \approx 0.233\text{V}$
- The inverter threshold voltage (midpoint): $V_M = \frac{V_{DD} - |V_{tp}| + V_{tn} \sqrt{\frac{\beta_n}{\beta_p}}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}} = \frac{3.3 - 0.8 + (0.75)\sqrt{1.67}}{1 + \sqrt{1.67}} = 1.51\text{V}$
- Note that: $V_{IL} < V_M < V_{IH}$: $1.26\text{V} < 1.51\text{V} < 1.7\text{V}$

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CMOS INVERTER

Transient Analysis

- Analyze Transient Characteristics of CMOS Gates by studying an Inverter
- Transient Analysis
 - Signal value as a function of time
- Transient Analysis of CMOS Inverter
 - $V_{in}(t)$, input voltage, function of time
 - $V_{out}(t)$, output voltage, function of time
 - V_{DD} and Ground, DC (not function of time)
 - find $V_{out}(t) = f(V_{in}(t))$
- Transient Parameters
 - Output signal rise (t_r) and fall time (t_f)
 - Propagation delay (t_p)



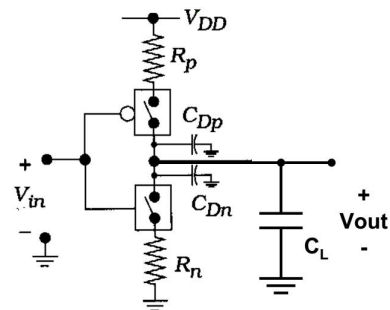
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CMOS INVERTER

Transient Response

- Response to step change in input
 - delays in output due to parasitic R & C
- Inverter RC Model
 - Resistances
 - $R_n = 1/[\beta_n(V_{DD} - V_{tn})]$
 - $R_p = 1/[\beta_p(V_{DD} - |V_{tp}|)]$
 - Output Capacitance (only output is important)
 - C_{Dn} (nMOS drain capacitance): $C_{Dn} = \frac{1}{2}C_{ox}W_nL + C_fA_{Dnbot} + C_{jsw}P_{Dnsw}$
 - C_{Dp} (pMOS drain capacitance): $C_{Dp} = \frac{1}{2}C_{ox}W_pL + C_fA_{Dpbot} + C_{jsw}P_{Dpsw}$
 - Load capacitance, due to gates attached at the output

$$C_L = N \times C_{in} = N \times (C_{Gn} + C_{Gp}), N \text{ is number of fan-out}$$
 - Total Output Capacitance: $C_{out} = C_{Dn} + C_{Dp} + C_L$



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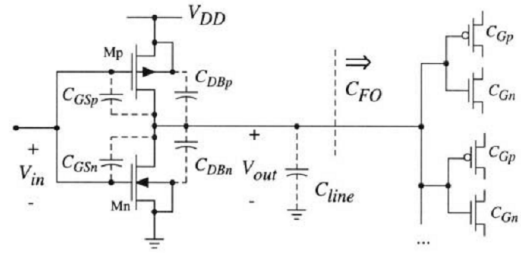
CMOS INVERTER

Capacitance

- Gate oxide capacitance: $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$
- The total gate capacitance of the nFET: $C_{Gn} = C_{ox}W_nL'$
- The total gate capacitance of the pFET: $C_{Gp} = C_{ox}W_pL'$
- The gate-drain capacitances of nFET and pFET

$$C_{GDn} = \frac{1}{2}C_{Gn} \quad C_{GDp} = \frac{1}{2}C_{Gp}$$

- The internal capacitance: $C_{int} = (C_{GDp} + C_{GDn} + C_{DBp} + C_{DBn})$
- Fan-out capacitance: $C_{FO} = FO \cdot (C_{Gn} + C_{Gp})$
- The external capacitance: $C_L = C_{FO} + C_{line}$
- The line capacitance (interconnect wiring): $C_{line} = \frac{\epsilon_{ox}}{t_{int}}Dw$ (D : distance, w : width, t_{int} : thickness of the oxide between the line and the substrate)
- The total capacitance: $C_{out} = (C_{GDp} + C_{GDn} + C_{DBp} + C_{DBn}) + (C_{FO} + C_{line}) = C_{int} + C_L$



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CMOS INVERTER

Capacitance

- Depletion capacitance (nFET):

$$C_{bot,n} = C_{j0n}A_{Dn}$$

$$C_{sw,n} = C_{jswn}P_{Dn}$$

- The drain-bulk capacitance (nFET):

$$C_{DBn} = K_{1/2}(0, V_{DD})C_{j0n}A_{Dn} + K_{1/3}(0, V_{DD})C_{jswn}P_{Dn}$$

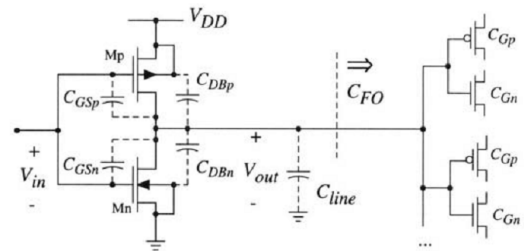
- Depletion capacitance (pFET):

$$C_{bot,p} = C_{j0p}A_{Dp}$$

$$C_{sw,p} = C_{jswp}P_{Dp}$$

- The drain-bulk capacitance (pFET):

$$C_{DBp} = K_{1/2}(0, V_{DD})C_{j0p}A_{Dp} + K_{1/3}(0, V_{DD})C_{jswp}P_{Dp}$$



$$K_{1/2}(0, V_{DD}) = \frac{2\phi_o}{V_{DD}} \left[\left(1 + \frac{V_{DD}}{\phi_o} \right)^{1/2} - 1 \right]$$

$$K_{1/3}(0, V_{DD}) = \frac{3\phi_o}{2V_{DD}} \left[\left(1 + \frac{V_{DD}}{\phi_{osw}} \right)^{2/3} - 1 \right]$$

A_{Dn} , P_{Dn} : bottom area and perimeter for nFET drain region

A_{Dp} , P_{Dp} : bottom area and perimeter for pFET drain region

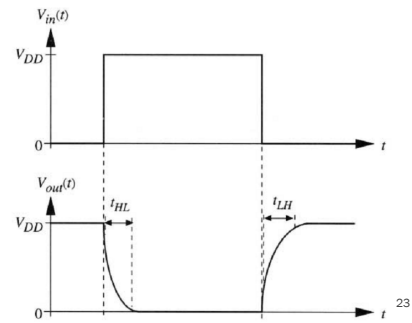
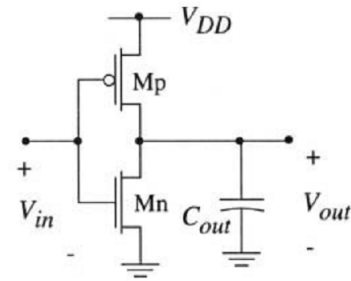
C_{j0} , C_{jsw} : zero-bias capacitance, zero-bias capacitance per unit length

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CMOS INVERTER

Switching Intervals

- When $V_{in} = V_{DD}$, nFET is ON, pFET is OFF $\rightarrow C_{out}$ discharges to 0V through Mn $\rightarrow$ Switching time is high-to-low time t_{HL} .
- t_{HL} is referred to fall time t_f .
- When $V_{in} = 0V$, nFET is OFF, pFET is ON $\rightarrow C_{out}$ charges to $V_{out} = V_{DD} \rightarrow$ Switching time is low-to-high time t_{LH} .
- t_{LH} is referred to rise time t_r .



CMOS INVERTER

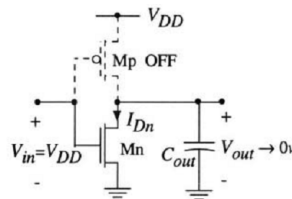
High-to-Low Time, t_{HL}

- Time for output to **fall** from '1' to '0'
- Derivation: $I_{Dn} = -C_{out} \frac{\partial V_{out}}{\partial t} = \frac{V_{out}}{R_n}$

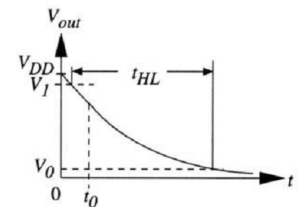
$$\frac{\beta_n}{2} (V_{DD} - V_{Tn})^2 = -C_{out} \frac{\partial V_{out}}{\partial t}$$

- At the beginning, Mn is saturated
- Initial condition, $V_{out}(t=0) = V_{DD}$
- Solution: $V_{out}(t) = (V_{DD} - V_{Tn}) \left[\frac{2e^{-(t-t_0)/\tau_n}}{1 + e^{-(t-t_0)/\tau_n}} \right]$

- Time constant for the discharge circuit: $\tau_n = \frac{C_{out}}{\beta_n (V_{DD} - V_{Tn})}$



(a) Discharge circuit



(b) Output voltage

CMOS INVERTER

High-to-Low Time, t_{HL}

- The value of t_{HL} is between 10% and 90% voltage: V_0 and V_1 respectively.

$$V_0 = 0.1V_{DD}, V_1 = 0.9V_{DD}$$

- t_{HL} can be computed using the integrals

$$t_{HL} = C_{out} \int_{V_{DD}-V_{Tn}}^{V_1} \frac{dV_{out}}{I_{Dn(sat)}} + C_{out} \int_{V_0}^{V_{DD}-V_{Tn}} \frac{dV_{out}}{I_{Dn(non-sat)}}$$

- Solve: $t_{HL} = s_n \tau_n$

where voltage-dependent scaling multiplier: $s_n = \frac{2(V_{Tn} - V_0)}{(V_{DD} - V_{Tn})} + \ln\left(\frac{2(V_{DD} - V_{Tn})}{V_0} - 1\right)$

- Time constant: $\tau_n = R_n C_{out}$

where R_n represents for nFET resistance: $R_n = \frac{1}{\beta_n(V_{DD} - V_{Tn})}$

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CMOS INVERTER

Fall Time, t_f

- Recall: Time for output to **fall** from '1' to '0'

- Definition:

- With the exponent decay, the time t_x for the voltage to fall from V_{DD} to V_x

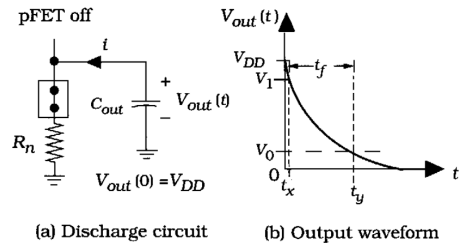
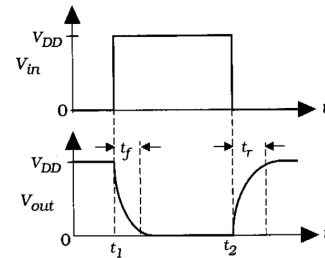
$$t_x = \tau_n \left[\ln\left(\frac{V_{DD}}{V_x}\right) \right]$$

- t_{HL} or t_f is time to fall from **90% value** [V_1, t_x] to **10% value** [V_0, t_y]

$$t_{HL} = t_{0.1} - t_{0.9}$$

$$t_f = t_{HL} = \tau_n \left[\ln\left(\frac{V_{DD}}{0.1V_{DD}}\right) - \ln\left(\frac{V_{DD}}{0.9V_{DD}}\right) \right] = \tau_n \ln[9]$$

- $t_f \approx 2.2\tau_n$



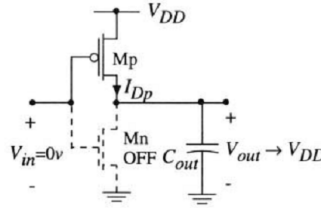
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CMOS INVERTER

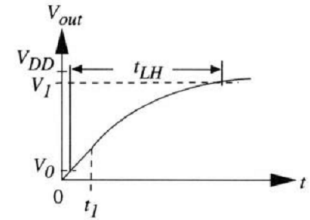
Low-to-High Time, t_{LH}

- Time for output to **rise** from '0' to '1'
- Derivation: $I_{Dp} = C_{out} \frac{\partial V_{out}}{\partial t}$

$$\frac{\beta_p}{2} (V_{DD} - |V_{Tp}|)^2 = C_{out} \frac{\partial V_{out}}{\partial t}$$



(a) Charge circuit



(b) Output voltage

- At the beginning, Mp is saturated
- Initial condition, $V_{out}(t=0) = V_0$
- Solution: $V_{out}(t) = V_{DD} - (V_{DD} - |V_{Tp}|) \left[\frac{2e^{-(t-t_0)/\tau_n}}{1 + e^{-(t-t_0)/\tau_n}} \right]$
- Time constant for the discharge circuit: $\tau_p = \frac{C_{out}}{\beta_p (V_{DD} - |V_{Tp}|)}$

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CMOS INVERTER

Low-to-High Time, t_{LH}

- t_{LH} as the time to charge C_{out} from V_0 (10%) to V_1 (90%) voltage

$$V_0 = 0.1V_{DD}, V_1 = 0.9V_{DD}$$

- Solve: $t_{LH} = s_p \tau_p$

where voltage-dependent scaling multiplier: $s_p = \frac{2(|V_{Tp}| - V_0)}{(V_{DD} - |V_{Tp}|)} + \ln \left(\frac{2(V_{DD} - |V_{Tp}|)}{V_0} - 1 \right)$

- Charging time constant: $\tau_p = R_p C_{out}$

where R_n represents for pFET resistance: $R_p = \frac{1}{\beta_p (V_{DD} - |V_{Tp}|)}$

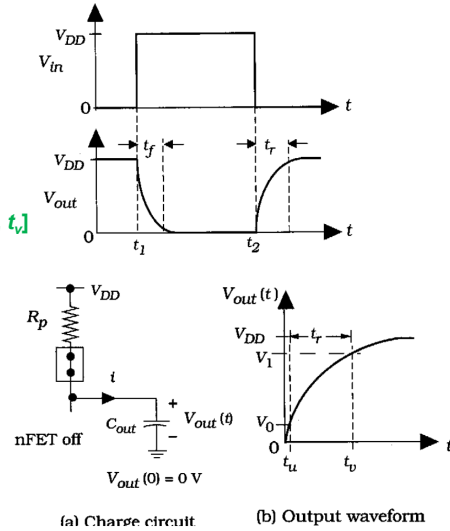
- Note that both R_n and R_p are inversely proportional to (W/L); increasing the aspect ratio thus decreases the equivalent resistance.

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CMOS INVERTER

Rise Time, t_r

- Time for output to **rise** from '0' to '1'
- Definition
 - t_{LH} or t_r is time to rise from 10% value $[V_0, t_u]$ to 90% value $[V_0, t_v]$
 - $t_r \approx 2.2\tau_p$

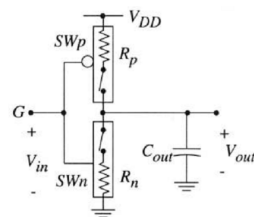


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CMOS INVERTER

RC Modelling

- The MOSFETs of an inverter are replaced by resistor-switch sub-networks.
- In the model, a cutoff transistor is represented by an open circuit, while an active MOSFET is represented by a closed switch in series with a parasitic drain-source resistance.
- The nFET equivalent resistance will be denoted by R_n , while R_p represents the pFET equivalent resistance.
- In this model, the logic circuit is based on the charging and discharging of C_{out} through the appropriate resistors.



(a) RC switching network

G	SW_p	SW_n
0	Closed	Open
1	Open	Closed

(b) Switching behavior

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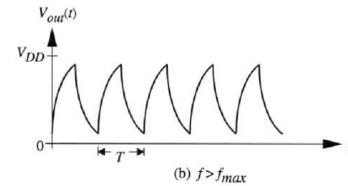
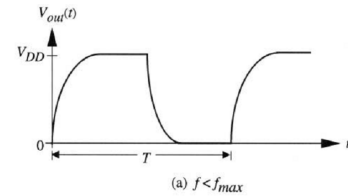
CMOS INVERTER

Maximum Switching Frequency

- The sum of the transient times ($t_{HL} + t_{LH}$) represents the minimum time needed for a gate to undergo a complete switching cycle, i.e., for the output to change from a logic 1 to a logic 0 voltage, and then back up to a logic 1 value.

- Maximum frequency:
$$f_{max} = \frac{1}{t_{HL} + t_{LH}}$$

- In Figure (a), for $f < f_{max}$, the output has sufficient time to react to changes in the inputs and exhibits proper form.
- In Figure (b), for $f > f_{max}$, the circuit does not have enough time to complete the charge or discharge event. This gives an output signal that has a limited amplitude that may cause a logic error.
- If $t_{HL} \approx 2.2\tau_n$, $t_{LH} \approx 2.2\tau_p$:
$$f_{max} \approx \frac{0.45}{\tau_n + \tau_p}$$



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CMOS INVERTER

Propagation Delay, t_p

- Measures speed of output reaction to input change

$$t_p = \frac{t_{pf} + t_{pr}}{2}$$

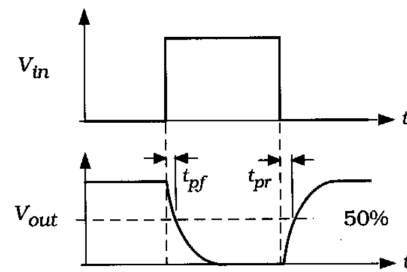
- Fall propagation delay, $t_{pf} = t_{0.5}$
 - time for output to fall by 50%
 - reference voltage by 50% V_{DD}

$$t_{pf} = \tau_n \left[\ln \left(\frac{V_{DD}}{0.5V_{DD}} \right) \right] \quad t_{pf} = \tau_n \ln[2]$$

- Rise propagation delay, t_{pr}
 - time for output to rise by 50%
 - reference voltage by 50% V_{DD}

$$t_{pr} = \tau_p \left[\ln \left(\frac{V_{DD}}{0.5V_{DD}} \right) \right] \quad t_{pr} = \tau_p \ln[2]$$

- Total Propagation Delay (Add rise and fall propagation delays for total value): $t_p \approx 0.35(\tau_n + \tau_p)$



Propagation delay measurement:
 - from time input reaches 50% value
 - to time output reaches 50% value

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CMOS INVERTER

Switching Speed - Resistance

- Rise and Fall Time: $t_r = 2.2\tau_p$, $t_f = 2.2\tau_n$
- Propagation Delay: $t_p = 0.35(\tau_n + \tau_p)$
- In general,
 - Delay $\propto \tau_n + \tau_p$
 - $\tau_n + \tau_p = C_{out}(R_n + R_p)$
- Define delay in terms of design parameters (if $V_t = V_{tn} = V_{tp}$)

$$R_n + R_p = \frac{(V_{DD} - V_t)(\beta_n + \beta_p)}{\beta_n\beta_p(V_{DD} - V_t)^2} = \frac{(\beta_n + \beta_p)}{\beta_n\beta_p(V_{DD} - V_t)}$$

- Beta Matched, if $\beta_n = \beta_p = \beta$: $R_n + R_p = \frac{2}{\beta(V_{DD} - V_t)} = \frac{2L}{\mu C_{ox}W(V_{DD} - V_t)}$
- Width Matched, if $W_n = W_p = W$ and $L_n = L_p = L$: $R_n + R_p = \frac{L(\mu_n + \mu_p)}{\mu_n\mu_p C_{ox}W(V_{DD} - V_t)}$

To decrease R's: $\downarrow L$, $\uparrow W$, $\uparrow V_{DD}$, ($\uparrow \mu_p$, $\uparrow C_{ox}$)

Provided by:

$$\tau_n = R_n C_{out}, \tau_p = R_p C_{out}$$

$$\beta = \mu C_{ox} \left(\frac{W}{L} \right)$$

$$R_n = 1/[\beta(V_{DD} - V_{tn})]$$

$$R_p = 1/[\beta(V_{DD} - V_{tp})]$$

$$C_{out} = C_{Dn} + C_{Dp} + C_L$$

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CMOS INVERTER

Switching Speed - Capacitance

- From Resistance, we have
 - $\downarrow L$, $\uparrow W$, $\uparrow V_{DD}$, ($\uparrow \mu_p$, $\uparrow C_{ox}$)
 - But $\uparrow V_{DD} \rightarrow$ increase power, $\uparrow W \rightarrow$ increase C_{out}
- Output capacitance C_{out}

$$C_{out} = \frac{1}{2C_{ox}L(W_n + W_p)} + C_j 2L(W_n + W_p) + 3C_{ox}L(W_n + W_p)$$

- assuming junction area $\sim W \cdot 2L$
- neglecting sidewall capacitance

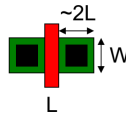
$$C_{out} \approx L(W_n + W_p) \left[3\frac{1}{2}C_{ox} + 2C_j \right]$$

$$C_{out} \propto L(W_n + W_p)$$

- To decrease C_{out} : $\downarrow L$, $\downarrow W$, ($\downarrow C_j$, $\downarrow C_{ox}$)

$$\text{Delay} \propto C_{out}(R_n + R_p) \propto LW \frac{L}{WV_{DD}} = \frac{L^2}{V_{DD}}$$

- Decreasing L (reducing feature size) is best way to improve speed!



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CMOS INVERTER

Switching Speed – Local modification

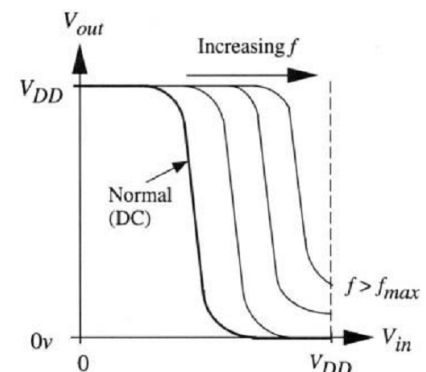
- Previous analysis applies to the overall design
 - shows that **reducing feature size** is critical for **higher speed**
 - general result useful for creating cell libraries
- How do you improve speed within a specific gate?
 - increasing W in one gate will not increase C_G of the load gates
 - $C_{out} = C_{Dn} + C_{Dp} + C_L$
 - increasing W in one logic gate will increase $C_{Dn/p}$ but not C_L
 - C_L depends on the size of the t_x gates at the output
 - as long as they keep minimum W , C_L will be constant
 - thus, **increasing W** is a good way to improve the speed within a local point
 - But, **increasing W increases chip area** needed, which is **bad**
 - fast circuits need more chip area (chip “real estate”)
 - **Increasing V_{DD}** is not a good choice because it **increases power consumption**

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CMOS INVERTER

Transient Effects on VTC

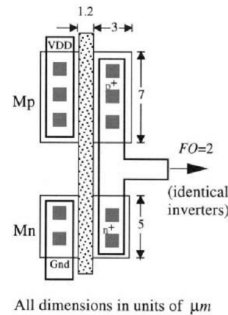
- VTC is defined to be the DC transfer curve, it is useful for illustrating the transient switching effects
- When the signal frequency is increased, the behavior of V_{out} as a function of V_{in} shows that there is a change in the response of the network.
- When the switching frequency exceeds then the circuit cannot respond to the quickly changing input.



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INVERTER DESIGN

- **Case study:** Consider the inverter layout in the figure below; all dimensions are in units of microns We will go through the complete calculation of the inverter characteristics with the information provided.



Oxide thickness: $x_{ox}=200\text{\AA}$

Gate overlap: $L_0=0.10\mu m$

Power supply: $V_{DD}=3.3v$

nFET Parameters

$V_{Tn} = +0.74 v, k_n' = 100 \mu A/V^2$

$C_{j0} = 2.82 \times 10^{-8} F/cm^2, \phi_0=90v, m_j=0.5$

$C_{jsw} = 4.62 \times 10^{-12} F/cm, \phi_{0sw}=95v, m_{jsw}=0.33$

pFET Parameters

$V_{Tp} = -0.90 v, k_p' = 38 \mu A/V^2$

$C_{j0} = 4.85 \times 10^{-8} F/cm^2, \phi_0=92v, m_j=0.5$

$C_{jsw} = 1.95 \times 10^{-12} F/cm, \phi_{0sw}=97v, m_{jsw}=0.33$

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INVERTER DESIGN

- **Case study: Inverter design**

- Since the draw channel is shown as $L' = 1.2\mu m$

- Gate overlap is $L_0 = 0.1\mu m$

- The electrical length is $L = L' - 2L_0 = 1 \mu m$

- The aspect ratios are $\left(\frac{W}{L}\right)_n = \left(\frac{5}{1}\right)$ $\left(\frac{W}{L}\right)_p = \left(\frac{7}{1}\right)$

- The device transconductance values

$$\beta_n = k_n' \left(\frac{W}{L}\right)_n = 500 \mu A/V^2 \quad \beta_p = k_p' \left(\frac{W}{L}\right)_p = 266 \mu A/V^2$$

- Inverter Threshold Voltage:

$$V_M = \frac{V_{DD} - |V_{Tp}| + \sqrt{\frac{\beta_n}{\beta_p}} V_{Tn}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}} = \frac{3.3 - 0.9 + \sqrt{\frac{500}{266}} 0.74}{1 + \sqrt{\frac{500}{266}}} = 1.44V$$

Less than one-half of power supply voltage as expected

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INVERTER DESIGN

- **Case study: Inverter design**

- Gate capacitance: (gate oxide thickness $t_{ox} = 200\text{\AA}$)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.854 \times 10^{-14})}{200 \times 10^{-8}} = 1.727 \times 10^{-7} \frac{F}{cm^2} = 1.727 \frac{fF}{\mu m^2}$$

- The total gate capacitance of the nFET

$$C_{Gn} = C_{ox}W_nL' = (1.727)(5)(1.2) = 10.362fF$$

- The total gate capacitance of the pFET

$$C_{Gp} = C_{ox}W_pL' = (1.727)(7)(1.2) = 14.507fF$$

- The gate-drain capacitances

$$C_{GDn} = \frac{1}{2}C_{Gn} = 5.181fF$$

$$C_{GDp} = \frac{1}{2}C_{Gp} = 7.253fF$$

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INVERTER DESIGN

- **Case study: Inverter design**

- Depletion capacitance (pFET):

$$C_{bot,p} = C_{j0}A_{Dp} = (0.485)(3.1)(7) = 10.525fF$$

$$C_{sw,p} = C_{jsw}P_{Dp} = (0.195)(2)(3.1 + 7) = 3.939fF$$

(We have $C_{j0} = 0.485fF/\mu m^2$, $C_{jsw} = 0.195fF/\mu m$)

- Averaging factors

$$K_{1/2}(0,3.3) = \frac{2(0.92)}{3.3} \left[\left(1 + \frac{3.3}{0.92} \right)^{1/2} - 1 \right] = 0.637$$

$$K_{1/3}(0,3.3) = \frac{3(0.97)}{2(3.3)} \left[\left(1 + \frac{3.3}{0.97} \right)^{2/3} - 1 \right] = 0.813$$

- The average value of the drain-bulk capacitance:

$$C_{DBp} = K_{1/2}(0,3.3)C_{bot} + K_{1/3}(0,3.3)C_{sw} = (0.637)(10.525) + (0.813)(3.939) = 9.907fF$$

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INVERTER DESIGN

- **Case study: Inverter design**

- Depletion capacitance (nFET):

$$C_{bot,n} = C_{j0}A_{Dn} = (0.282)(3.1)(5) = 4.371fF$$

$$C_{sw,n} = C_{jsw}P_{Dn} = (0.462)(2)(3.1 + 5) = 7.484fF$$

(We have $C_{j0} = 0.282fF/\mu m^2$, $C_{jsw} = 0.462fF/\mu m$)

- Averaging factors

$$K_{1/2}(0,3.3) = \frac{2(0.90)}{3.3} \left[\left(1 + \frac{3.3}{0.90} \right)^{1/2} - 1 \right] = 0.633$$

$$K_{1/3}(0,3.3) = \frac{3(0.95)}{2(3.3)} \left[\left(1 + \frac{3.3}{0.95} \right)^{1/3} - 1 \right] = 0.741$$

- The average value of the drain-bulk capacitance:

$$C_{DBn} = K_{1/2}(0,3.3)C_{bot} + K_{1/3}(0,3.3)C_{sw} = (0.633)(4.371) + (0.741)(7.484) = 8.312fF$$

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INVERTER DESIGN

- **Case study: Inverter design**

- The internal capacitance

$$C_{int} = (C_{GDP} + C_{GDn} + C_{DBp} + C_{DBn}) = (7.253 + 5.181 + 9.907 + 8.312) = 30.653fF$$

- Fan-out capacitance :

$$C_{FO} = 2(C_{Gn} + C_{Gp}) = 2(10.362 + 14.507) = 49.738fF$$

- Interconnect: The figure does not provide any information on the interconnect line capacitance → can ignore ($C \approx 0$). However, for complete calculation, let us assume that the metal interconnect runs over a field oxide that has a thickness of $0.6\mu m$.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{int}} = \frac{(3.9)(8.854 \times 10^{-14})}{0.6 \times 10^{-4}} = 5.76 \times 10^{-9} \frac{F}{cm^2}$$

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INVERTER DESIGN

- **Case study: Inverter design**

- The metal line is approximately $2\mu\text{m}$ wide so we will guess a length of $50\mu\text{m}$ as being reasonable.

$$C_{line} = (5.76 \times 10^{-9}) (2 \times 10^{-4})(50 \times 10^{-4}) = 5.755fF$$

- The external capacitance C_L is obtained by

$$C_L = C_{FO} + C_{line} = 49.738 + 5.755 = 55.493fF$$

- The total capacitance

$$C_{out} = C_{int} + C_L = 30.738 + 55.493 = 86.146fF$$

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INVERTER DESIGN

- **Case study: Inverter design**

- Time constants (nFET):

$$R_n = \frac{1}{\beta_n(V_{DD} - V_{Tn})} = \frac{1}{500 \times 10^{-6}(3.3 - 0.74)} = 781.25\Omega$$

$$\tau_n = R_n C_{out} = (781.25)(86.146 \times 10^{-15}) = 67.302ps$$

- Time constants (pFET):

$$R_p = \frac{1}{\beta_p(V_{DD} - V_{Tp})} = \frac{1}{266 \times 10^{-6}(3.3 - 0.9)} = 1566.42\Omega$$

$$\tau_p = R_p C_{out} = (1566.42)(86.146 \times 10^{-15}) = 134.94ps$$

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INVERTER DESIGN

Case study: Inverter design

- Switching Time $t_{HL} = s_n \tau_n$ $t_{LH} = s_p \tau_p$

- The nFET voltage multiplier for the high-to-low time, we use 10% voltage $V_0 = 0.1 V_{DD} = 0.33V$

$$s_n = \frac{2(V_{Tn} - V_0)}{(V_{DD} - V_{Tn})} + \ln \left(\frac{2(V_{DD} - V_{Tn})}{V_0} - 1 \right) = \frac{2(0.74 - 0.33)}{(3.3 - 0.74)} + \ln \left(\frac{2(3.3 - 0.74)}{0.33} - 1 \right) = 2.995$$

- The pFET multiplier for the low-to-high time

$$s_p = \frac{2(|V_{Tp}| - V_0)}{(V_{DD} - |V_{Tp}|)} + \ln \left(\frac{2(V_{DD} - |V_{Tp}|)}{V_0} - 1 \right) = \frac{2(0.9 - 0.33)}{(3.3 - 0.9)} + \ln \left(\frac{2(3.3 - 0.9)}{0.33} - 1 \right) = 3.082$$

- The switching time: $t_{HL} = s_n \tau_n = (2.995)(67.302 \times 10^{-12}) = 201.57ps$

$$t_{LH} = s_p \tau_p = (3.082)(134.940 \times 10^{-12}) = 415.89ps$$

- The maximum frequency: (represents the upper limit for the single circuit)

$$f_{max} = \frac{1}{t_{HL} + t_{LH}} = 1.62GHz$$

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INVERTER DESIGN

Case study: Inverter design

- Since rail-to-rail output voltage levels are automatic from the topology of the circuit, the design is directed towards either shaping the voltage transfer curve or establishing the necessary transient switching times. This is accomplished by specifying the device parameters

$$\beta_n = k'_n \left(\frac{W}{L} \right)_n \quad \beta_p = k'_p \left(\frac{W}{L} \right)_p$$

- Choosing the aspect ratios establishes both the DC and the transient switching times. As we will see below, the ratio is chosen to set the inverter midpoint voltage while the individual values of and are adjusted according to the needed switching times.

- Inverter Threshold Voltage (V_I or V_M): The value of V_I can be set by the designer by adjusting the nFET/pFET device transconductance ratio (β_n / β_p)

$$V_I = V_M = \frac{V_{DD} - |V_{Tp}| + \sqrt{\frac{\beta_n}{\beta_p} V_{Tn}}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}} \quad \frac{\beta_n}{\beta_p} = \left(\frac{V_{DD} - V_I - |V_{Tp}|}{V_I - V_{Tn}} \right)^2$$

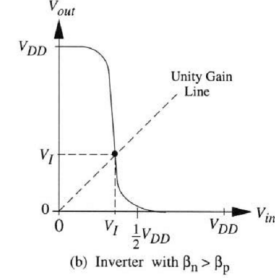
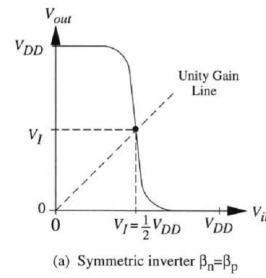
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INVERTER DESIGN

Case study: Inverter design

- The ratio (β_n/β_p) for the desired value of V_I

$$\frac{\beta_n}{\beta_p} = \left(\frac{V_{DD} - V_I - |V_{Tp}|}{V_I - V_{Tn}} \right)^2$$



- A symmetrical CMOS Inverter is defined to have (figure (a))

$$V_I = \frac{1}{2} V_{DD} \quad V_I = \frac{V_{DD} - |V_{Tp}| + \sqrt{1} V_{Tn}}{1 + \sqrt{1}} = \frac{1}{2} V_{DD}$$

- The switch point is at one-half the power supply voltage. Assuming that the process is polarity-symmetric with $V_{Tn} = |V_{Tp}| = V_T$ this condition can be achieved by designing the circuit with $\beta_n = \beta_p$
- In terms of the aspect ratios, this requires that $\left(\frac{W}{L}\right)_p = \frac{k'_n}{k'_p} \left(\frac{W}{L}\right)_n$
- Since $k'_n > k'_p$, this design requires that the pFET aspect ratio is larger than the nFET aspect ratio by a factor ($k'_n/k'_p \approx 2.5$) and $(W/L)_n = (W/L)_p \rightarrow V_I < V_{DD}/2$ (figure (b))

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INVERTER DESIGN

Case study: Inverter design

- In general case:

$$\frac{\beta_n}{\beta_p} = \frac{k'_n \left(\frac{W}{L}\right)_n}{k'_p \left(\frac{W}{L}\right)_p}$$

- If the ratio of (β_n/β_p) > 1 , then V_I is moved to the left of ($V_{DD}/2$), $V_I < V_{DD}/2$
- If the ratio of (β_n/β_p) < 1 , then $V_I > V_{DD}/2$ (less-likely case)
- In practice, the actual values of the aspect ratio may be chosen by the layout designer in non-critical datapath circuits simply to fit in the allocated area.

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INVERTER DESIGN

Case study: Inverter design

- Example: Design an inverter with the power supply $V_{DD} = 3.3\text{V}$, and the device parameters are $V_{Tn} = 0.74\text{V}$, $V_{Tp} = -0.9\text{V}$, $k'_n = 100\mu\text{A/V}^2$, $k'_p = 38\mu\text{A/V}^2$

- Solution: Suppose we want to design an inverter with $V_I = V_{DD}/2 = 1.65\text{V}$

$$\frac{\beta_n}{\beta_p} = \left(\frac{V_{DD} - V_I - |V_{Tp}|}{V_I - V_{Tn}} \right)^2 = \left(\frac{3.3 - 1.65 - 0.9}{1.5 - 0.74} \right)^2 = 0.97$$

- Relative device sizes

$$\frac{\beta_n}{\beta_p} = \frac{k'_n \left(\frac{W}{L} \right)_n}{k'_p \left(\frac{W}{L} \right)_p} \leftrightarrow \frac{100 \left(\frac{W}{L} \right)_n}{38 \left(\frac{W}{L} \right)_p} = 0.97$$

$$\left(\frac{W}{L} \right)_p = 2.7 \left(\frac{W}{L} \right)_n$$

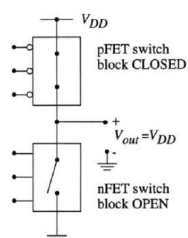
- This has the expected results that the pFET must be larger than the nFET

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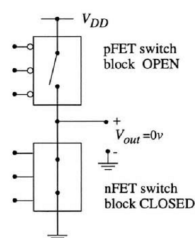
CMOS NAND GATES

- NAND 2-input (2 pFETs in parallel, 2 nFETs in series)

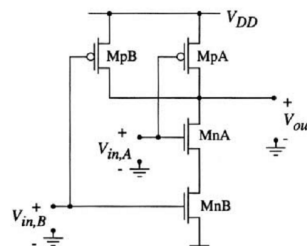
- Equation: $f = \overline{A \cdot B} = \bar{A} + \bar{B}$



(a) Logic 1 output



(b) Logic 0 output



(a) Logic circuit

$V_{in,A}$	$V_{in,B}$	V_{out}
0v	0v	V_{DD}
0v	V_{DD}	V_{DD}
V_{DD}	0v	V_{DD}
V_{DD}	V_{DD}	0v

(b) Operation summary



(a) Symbol

A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

(b) Truth table

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CMOS NAND GATES

■ NAND 2-input

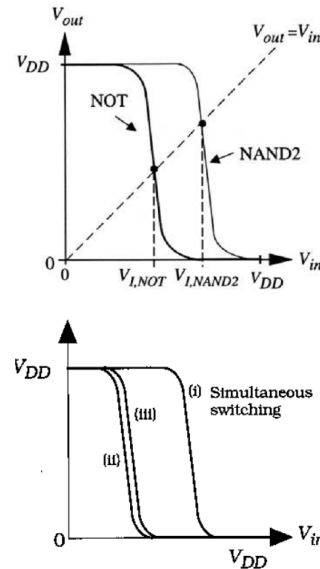
- Calculate V_M for NAND:

($V_M = V_A = V_B = V_{out}$, set $I_{Dn} = I_{Dp}$, solve for V_M)

$$V_M = \frac{V_{DD} - |V_{Tp}| + \frac{1}{2} \sqrt{\frac{\beta_n}{\beta_p}} V_{Tn}}{1 + \frac{1}{2} \sqrt{\frac{\beta_n}{\beta_p}}}$$

- denominator reduced more $\rightarrow$ VTC shifts right

but, since $\mu_n > \mu_p$, $V_M \approx V_{DD}/2$ when $W_n = W_p$



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CMOS NAND GATES

■ NAND 2-input

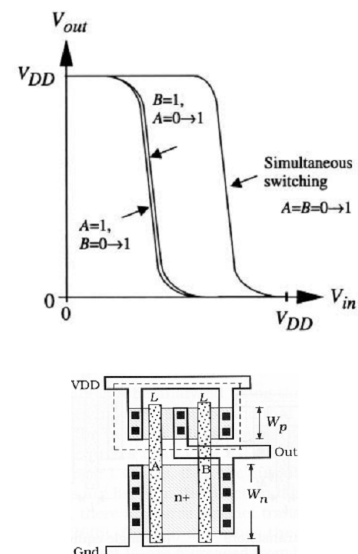
- 0,0 to 1,1 state transitions

- all transistors change states (on, off)

- The DC transfer characteristics depend on the input combinations. There are three input combinations that result in the output voltage changing from a high state to a low state. The possibilities are:

- $V_{in,A} = V_{in,B}$ are simultaneously switched from 0V towards VDD
 - $V_{in,A} = V_{DD}$ while $V_{in,B}$ is switched from 0V towards VDD
 - $V_{in,B} = V_{DD}$ while $V_{in,A}$ is switched from 0V towards VDD

- Series nMOS means more resistance to output **falling**, shifts VTC to right $\rightarrow$ To balance this effect and set V_M to $V_{DD}/2$, can increase β by increasing W_n



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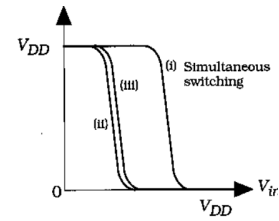
CMOS NAND GATES

DC Analysis

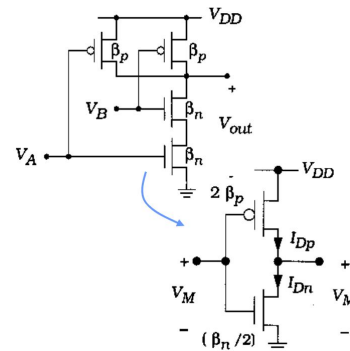
- Multiple Inputs
- Multiple Transitions
- Multiple VTCs
 - VTC varies with transition
 - transition from 0,0 to 1,1 pushed right of others
 - V_M varies with transition
 - assume all transistors have same L
 - $V_M = V_A = V_B = V_{out}$
 - if $W_{pA} = W_{pB}$ and $W_{nA} = W_{nB}$
 - series nMOS, $\beta_N \Rightarrow \frac{1}{2}\beta_n$
 - parallel pMOS, $\beta_P \Rightarrow 2\beta_p$

	V_A	V_B	V_{out}
(i)	0	0	V_{DD}
(ii)	0	V_{DD}	V_{DD}
(iii)	V_{DD}	0	V_{DD}
	V_{DD}	V_{DD}	0

(a) Transition table



(b) VTC family

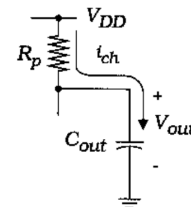
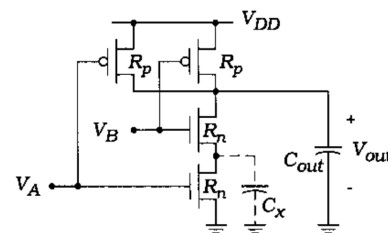


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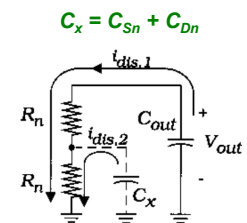
CMOS NAND GATES

Transient Analysis

- NAND RC Circuit
 - R: standard channel resistance
 - C: $C_{out} = C_L + C_{Dn} + 2C_{Dp}$
- Rise Time, t_r
 - Worst case charge circuit: 1 pMOS ON
 - $t_r = 2.2\tau_p$ where $\tau_p = R_p C_{out}$
 - best case charge circuit: 2 pMOS ON, $R_p \rightarrow R_p/2$
- Fall Time, t_f
 - Discharge Circuit
 - 2 series nMOS, $R_n \rightarrow 2R_n$
 - must account for internal cap, C_x
 - $t_f = 2.2\tau_n$ where $\tau_n = C_{out}(2R_n) + C_x R_n$



(a) Charging circuit



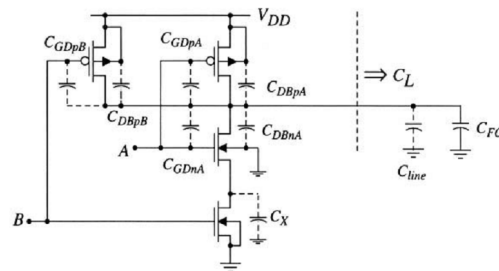
(b) Discharging circuit

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CMOS NAND GATES

■ NAND 2-input

- Output capacitance: $C_{out} = C_{FET} + C_L$
- MOSFET Parasitics: $C_{FET} = C_{GDnA} + C_{GDpA} + C_{GDpB} + C_{DBnA} + C_{DBpA} + C_{DBpB}$
- External load capacitance : $C_L = C_{line} + C_{FO}$
- Internal node capacitance: $C_X = C_{GSnA} + C_{GSnB} + C_{n+}$
- C_{n+} is the total capacitance of the n+ drain/source region between the series-connected nFETs.

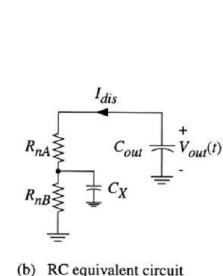
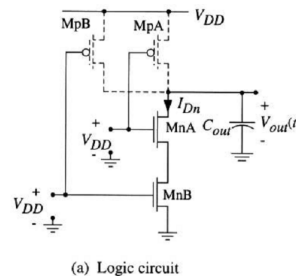
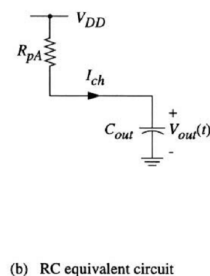
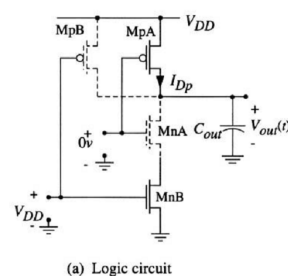


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CMOS NAND GATES

■ NAND 2-input

- **Output charge time:** If either A or B (or both) switch to logic 0 values, charges through the appropriate pFET transistors
- The low-to-high time: $t_{LH} = s_p \tau_p$
- Where $\tau_p = R_{pA} C_{out}$



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CMOS NAND GATES

- **NAND 2-input**
- **Output Discharge Delay Time:** Discharging occurs when both A and B increase to logic 1 voltages. MOSFETs M_{nA} and M_{nB} are both active and provide a conducting path between C_{out} and ground
- The high-to-lo time: $t_{HL} = S_n \tau_n$
- Where $\tau_n = (R_{nA} + R_{nB})C_{out} + R_{nA}C_X$
- To estimate the high-to-low time, we are modelling the output voltage as an exponential of the form
$$V_{out}(t) = V_{DD}e^{-t/\tau_n} \quad t \approx \tau_n \ln \left[\frac{V_{DD}}{V_{out}(t)} \right]$$
- t_{HL} is defined as the time required for the voltage to fall from the 90% voltage to the 10% voltage, we have that
$$t_{HL} \approx \tau_n \ln[9] \approx 2.2\tau_n$$
- The output capacitance must discharge through the series-connected nFETs, which implies that the value of t_{HL} can be large.

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CMOS NAND GATES

- **NAND 2-input**
- Design: pFETs are in parallel, nFETs are in series. The device sizes establish the DC critical voltages such as switching voltage and directly determine the transient time intervals.
- We design both pFETs to be same size $(W/L)_p$, and both nFETs to be same size $(W/L)_n$.
- The device sizes establish the DC critical voltages such as switching voltage and directly determine the transient time intervals.
- Firstly, adjust the ratio (β_n / β_p) , and then choose the aspect ratios according to the technology limits and the desired transient response.
- If timing is critical, we may choose the aspect ratios needed to satisfy the switching response and only calculate the DC characteristics as an after thought.

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CMOS NAND GATES

■ NAND 2-input

- The low-to-high time t_{LH} is controlled by the pFET aspect ratios $(W/L)_{pA}$ and $(W/L)_{pB}$
- We can design both pFETs to be the same size $(W/L)_p$ such that either transistor can individually meet the rise time specification.

$$\left(\frac{W}{L}\right)_p = \frac{C_{out}}{\tau_p k'_p (V_{DD} - |V_{Tp}|)}$$

- The series-connected nFETs limit the discharging response.

$$\left(\frac{W}{L}\right)_n = \frac{2C_{out} + C_x}{\tau_n k'_n (V_{DD} - V_{Tn})} \quad \tau_n \approx R_n (2C_{out} + C_x) \quad (\text{for identical nFETs})$$

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CMOS NAND GATES

■ NAND N-input

- The NAND-2 structure can be extended to an N-input NAND gate by using N complementary pairs where the nFETs are in series and the pFETs are in parallel.

$$V_I = \frac{V_{DD} - |V_{Tp}| + \frac{1}{N} \sqrt{\frac{\beta_n}{\beta_p}} V_{Tn}}{1 + \frac{1}{N} \sqrt{\frac{\beta_n}{\beta_p}}}$$

- For the case $N=1$, this is identical to the inverter results.
- Multiple-input NAND gates are easily implemented in circuit and logic designs. However, the output capacitance increases with N due to the pMOS parasitics, slowing down the overall response.
- The worst-case problem is that the discharge time is limited by the discharge through the nFET chain that consists of series-connected transistors.
- These considerations generally limit N to a maximum of 3 or 4 inputs in realistic designs

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CMOS NAND GATES

■ NAND Design

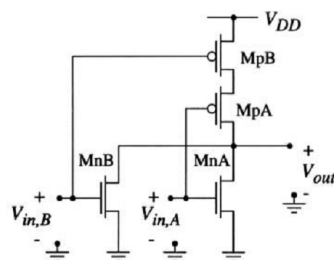
- The design process starts by
 - Estimating the capacitance values
 - Designing the network (NMOS and PMOS network) around these choices
 - A computer simulation is then used to check the actual behavior with a specified external load.
- Another approach
 - Choose reasonable values for the device sizes
 - Simulate the circuit
 - Use the equations to adjust the aspect ratios as needed for the transient response.

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CMOS NOR GATES

■ NOR 2-input (2 pFETs in series, 2 nFETs in parallel)

- Equation: $f = \overline{A + B} = \bar{A} \cdot \bar{B}$



(a) Logic circuit

$V_{in,A}$	$V_{in,B}$	V_{out}
0v	0v	V_{DD}
0v	V_{DD}	0v
V_{DD}	0v	0v
V_{DD}	V_{DD}	0v

(b) Operation summary

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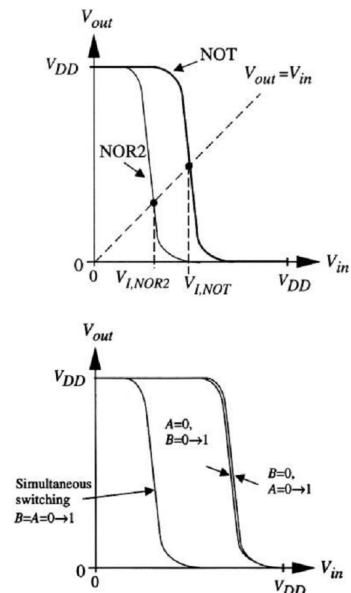
CMOS NOR GATES

NOR 2-input

- Calculate V_M for NOR:

$$V_M = \frac{V_{DD} - |V_{Tp}| + 2 \sqrt{\frac{\beta_n}{\beta_p}} V_{Tn}}{1 + 2 \sqrt{\frac{\beta_n}{\beta_p}}}$$

- series pMOS resistance means slower rise
 - VTC shifted to the left
 - to set V_M to $V_{DD}/2$, increase $W_p \rightarrow$ this will increase β_p

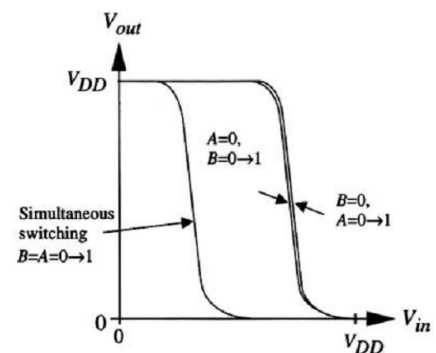


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CMOS NOR GATES

NOR 2-input

- The DC voltage switching characteristics of a NOR gate depends on how the inputs are changed. Three different curves are shown in the VTC of the figure, each representing a different switching combination.
 - $V_{in,A} = V_{in,B}$ are simultaneously switched from 0V towards VDD
 - $V_{in,A} = V_{DD}$ while $V_{in,B}$ is switched from 0V towards VDD
 - $V_{in,B} = V_{DD}$ while $V_{in,A}$ is switched from 0V towards VDD
- This is similar to the situation encountered with the NAND2 gate, except that the individual switching events are on the right side of the VTC, which is opposite to behavior of the NAND2 gate.



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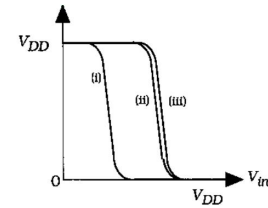
CMOS NOR GATES

DC Analysis

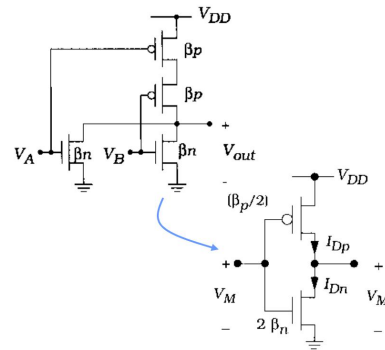
- Similar analysis to NAND
- Critical Transition
 - transition from 0,0 to 1,1 and when all transistors change
- For NOR 2-input with critical transition
 - if $W_{pA} = W_{pB}$ and $W_{nA} = W_{nB}$
 - parallel nMOS, $\beta_N \Rightarrow 2\beta_n$
 - series pMOS, $\beta_P \Rightarrow \frac{1}{2}\beta_p$

	V_A	V_B	V_{out}
	0	0	V_{DD}
(iii)	0	V_{DD}	0
(ii)	V_{DD}	0	0
(i)	V_{DD}	V_{DD}	0

(a) Transition table



(b) VTC family

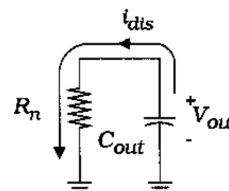
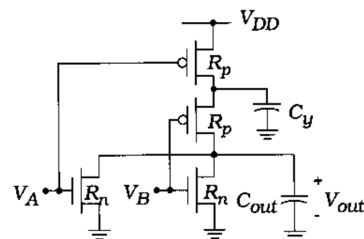


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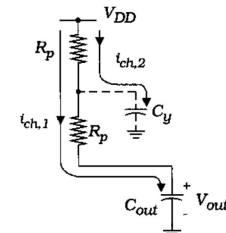
CMOS NOR GATES

Transient Analysis

- NOR RC Circuit
 - R: standard channel resistance
 - C: $C_{out} = C_L + 2C_{Dn} + C_{Dp}$
- Fall Time, t_f
 - Worst case charge circuit: 1 nMOS ON
 - $t_f = 2.2\tau_n$ where $\tau_n = R_n C_{out}$
 - best case charge circuit: 2 nMOS ON, $R_n \rightarrow R_n/2$
- Rise Time, t_r
 - Charge Circuit
 - 2 series pMOS, $R_p \rightarrow 2R_p$
 - must account for internal cap, C_y
 - $t_r = 2.2\tau_p$ where $\tau_p = C_{out}(2R_p) + C_y R_p$



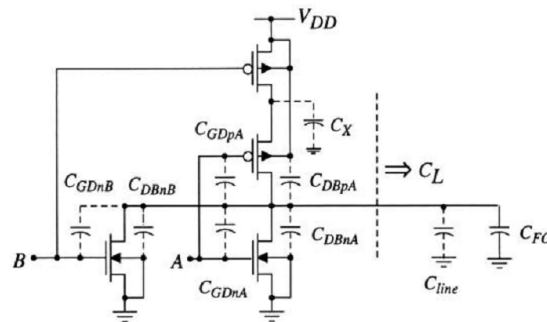
$$C_y = C_{Sp} + C_{Dp}$$



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CMOS NOR GATES

- **NOR 2-input**
- Output capacitance: $C_{out} = C_{FET} + C_L$
- MOSFET Parasitics: $C_{FET} = C_{GDnA} + C_{GDnB} + C_{GDpA} + C_{DBnA} + C_{DBnB} + C_{DBpA}$
- External load capacitance : $C_L = C_{line} + C_{FO}$



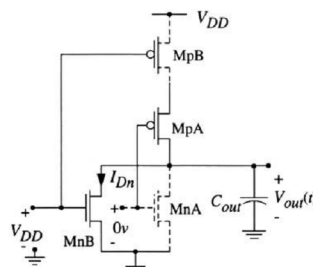
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CMOS NOR GATES

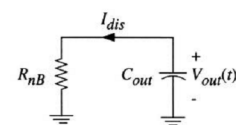
- **NOR 2-input**
- **Output Discharge Time:**
- The high-to-low time t_{HL} is computed by noting that C_{out} discharges through the nFET MnA or MnB.

$$t_{HL} = s_n \tau_n$$

$$\tau_n = \frac{C_{out}}{\beta_n (V_{DD} - V_{Tn})}$$



(a) Logic circuit



(b) RC equivalent circuit

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CMOS NOR GATES

■ NOR 2-input

- **Output Charge Delay Time:** With both A and B at logic 0 levels, pFETs MpA and MpB provide a conducting path between the output and the power supply
- Constructing the RC model gives a time constant for the discharge event as

$$\tau_p = (R_{pA} + R_{pB})C_{out} + R_{pB}C_X$$

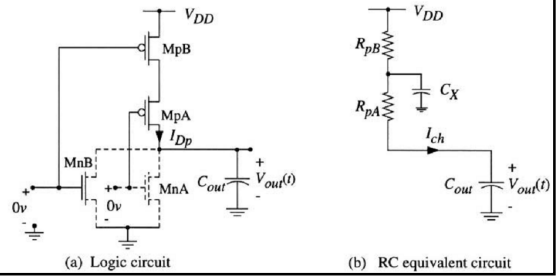
- The capacitance between the two series-connected p-channel transistors: $C_X = C_{GSpA} + C_{GSpB} + C_{p+}$ (C_{p+} is the total depletion capacitance between the two.)
- The low-to-high time t_{LH} is estimated using exponential

$$t_{LH} = s_p \tau_p$$

$$V_{out}(t) = V_{DD} e^{-t/\tau_p}$$

$$t \approx \tau_p \ln \left[\frac{V_{DD}}{V_{out}(t)} \right]$$

$$t_{LH} \approx \tau_p \ln[9] \approx 2.2\tau_p$$



CMOS NOR GATES

■ NOR 2-input

- Design: pFETs are in series, nFETs are in parallel. The choice of aspect ratios affects the gate threshold voltage and the transient characteristics, as is the case for all static CMOS gates.
- We design both pFETs to be same size $(W/L)_p$, and both nFETs to be same size $(W/L)_n$.
- NOR gate design is similar to that discussed for NAND circuits. To set the value of V_I we adjust the ratio (β_n/β_p) , the transient specifications dictate the specific values of the aspect ratios.
- The high-to-low time t_{HL} is set by the values $(W/L)_{nA}$ and $(W/L)_{nB}$. Since the two are in parallel, the simplest choice is to calculate the aspect ratio

$$\left(\frac{W}{L}\right)_n = \frac{C_{out}}{\tau_n k'_n (V_{DD} - V_{Tn})}$$

- The charging event is controlled by the series-connected pFETs

$$\left(\frac{W}{L}\right)_p = \frac{2C_{out} + C_X}{\tau_p k'_p (V_{DD} - |V_{Tp}|)} \quad \tau_p \approx R_p (2C_{out} + C_X) \quad (\text{for identical transistors})$$

CMOS NOR GATES

- **NOR N-input**
- An N-input NOR gate can be constructed by using complementary structuring with N nFETs in parallel, and N pFETs in series.

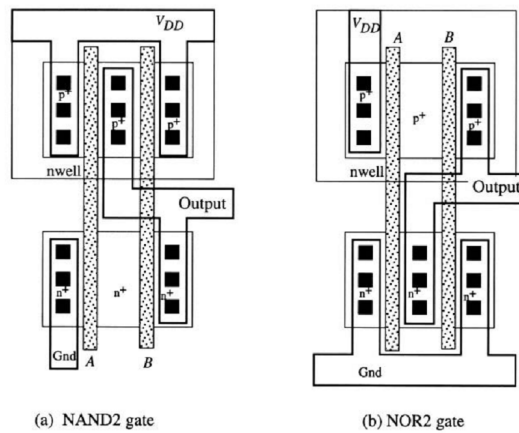
$$V_I = \frac{V_{DD} - |V_{Tp}| + N \sqrt{\frac{\beta_n}{\beta_p}} V_{Tn}}{1 + N \sqrt{\frac{\beta_n}{\beta_p}}}$$

- Note that the case where N=1 is identical to the inverter.
- NOR gate logic is very popular.
- However, it also has an output capacitance problem due to the contributions from the parallel nMOS devices, and requires charging through a pMOS chain which limits the transient response.
- In practice, we limit N to a maximum of 2 or 3, unless speed is not an issue.

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NAND/NOR PERFORMANCE

- **Layout of NAND 2-input and NOR 2-input**



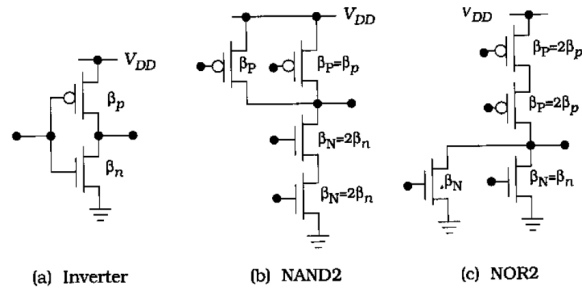
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NAND/NOR PERFORMANCE

Transient Analysis

- Inverter: symmetry ($V_M = V_{DD}/2$), $\beta_n = \beta_p$
 - $(W/L)_p = \mu_n/\mu_p(W/L)_n$
- Match INV performance with NAND
 - pMOS, $\beta_p = \beta_p$, same as inverter
 - nMOS, $\beta_n = 2\beta_n$, to balance for 2 series nMOS
- Match INV performance with NOR
 - pMOS, $\beta_p = 2\beta_p$, to balance for 2 series pMOS
 - nMOS, $\beta_n = \beta_n$, same as inverter
- NAND and NOR will still be slower due to larger C_{out}
- This can be extended to 3, 4, ... N-input NAND/NOR gates

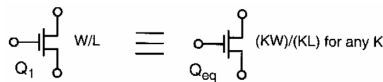
β is adjusted by changing transistor size (width)



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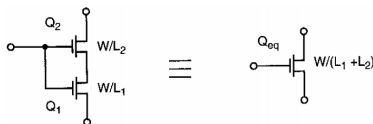
SERIES/PARALLEL EQUIVALENT CIRCUITS

- Scale both W and L
 - no effective change in W/L
 - increases gate capacitance



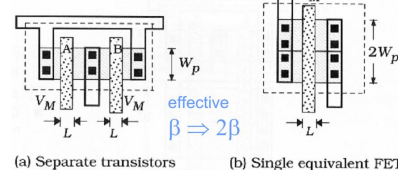
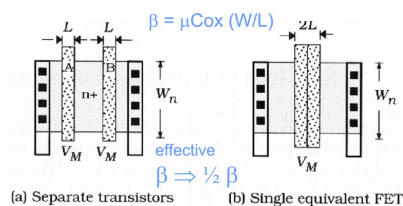
Series Transistors

- increases effective L



Parallel Transistors

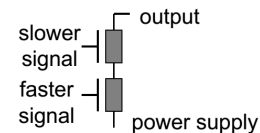
- increases effective W



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PERFORMANCE CONSIDERATIONS

- Speed based on β_n , β_p and parasitic capacitors
- DC performance (V_M , noise) based on β_n/β_p
- Design for speed not necessarily provide good DC performance
- Generally, set transistor size to optimize speed and then test DC characteristics to ensure adequate noise immunity
- Review Inverter: Our performance reference point
 - for symmetry ($V_M = V_{DD}/2$), $\beta_n = \beta_p$
 - which requires $(W/L)_p = \mu_n/\mu_p(W/L)_n$
- Use inverter as **reference point** for more **complex logic gates**
- Apply slowest arriving inputs to series node closest to output
 - let faster signals begin to charge/discharge nodes closer to V_{DD} and Ground



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CMOS COMPLEX LOGIC GATES

- Complex logic gates provide functions that consist of several primitive NOT, AND, or OR operations. Consider a three-variable function with input variables a, b and c. Suppose that we construct the function:

$$f(a, b, c) = \overline{a \cdot \bar{b} \cdot c + \bar{a} \cdot b \cdot c + a \cdot b \cdot c}$$

- This is a canonical AOI (AND-OR-INVERTER) equation.
- If we reverse the ordering of the logic operations (replace $\cdot$ with $+$, $+$ with $\cdot$), we obtain an OAI (OR-AND-INVERTER) expression.

$$g(a, b, c) = \overline{(a + \bar{b} + c) \cdot (\bar{a} + b + c) \cdot (a + b + c)}$$

- Complex logic gates are constructed using the CMOS inverter as a basis.
- In order to construct a complex logic gate, let us replace the single inverter nFET by an array of nFETs that are connected to operate as a large switch.
 - NAND Gates: pFETs are in parallel, nFETs are in series
 - NOR Gates: pFETs are in series, nFETs are in parallel
- Similarly, we will substitute an array of pFETs for the single pFET used in the inverter, and view the pFET array as a "giant" switch.

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CMOS COMPLEX LOGIC GATES

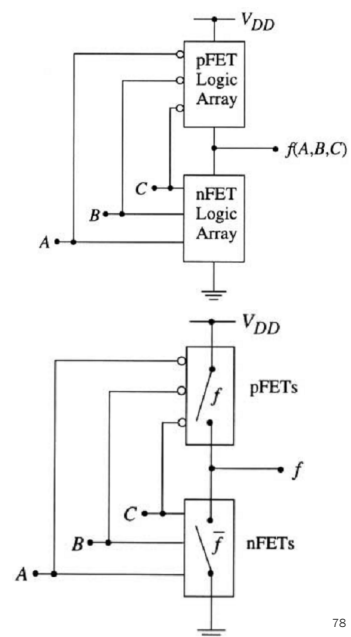
- The operation of the nFET array complements the operation of the pFET array. This means that if one array is a closed switch, the other must be open.
- The general structure of a complex logic gate can be created by the following steps:
 - Provide a complementary pair (an nFET and a pFET with a common gate) for each input;
 - Replace the single nFET with an **array** of nFETs that connects the output to ground;
 - Replace the single pFET with an **array** of pFETs that connects the output to V_{DD} ;
 - Design the nFET and pFET switching network so that only one network acts as a closed switch for any given input combination.

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CMOS COMPLEX LOGIC GATES

The general complex logic gate circuit

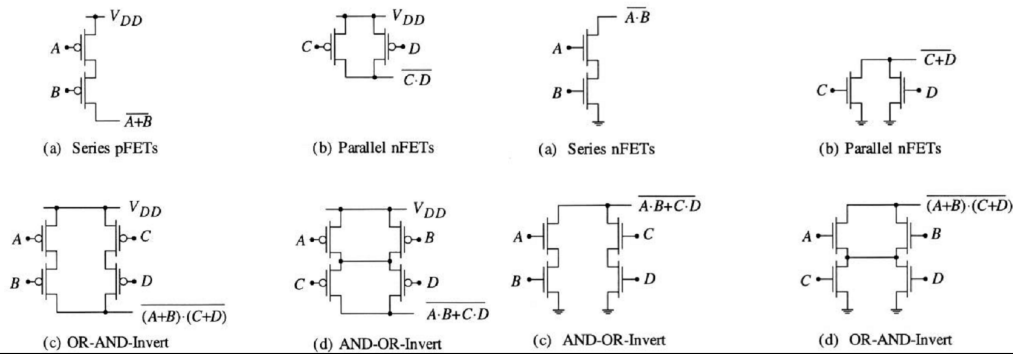
- The use of complementary pairs requires that every input drives both an nFET and a pFET. Complex logic functions can be implemented by designing the nFET and pFET switching arrays such that only one composite switch is closed for a given set of inputs.
- If the nFET switch is closed while the pFET switch is open, then the output is a logic 0. Conversely, a closed pFET switch and an open nFET switch results in a logic 1 output.
- The arrays must be designed to **avoid two situations**:
 1. both switches are open at the same time, since this gives an undetermined value;
 2. both are closed at the same time, since the voltage will not be a well-defined logic voltage.



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CMOS COMPLEX LOGIC GATES

- Recall that connecting nFETs in series gives the AND-NOT function, while wiring nFETs in parallel results in the OR-NOT operation;
- In other words, the logic formation rules also apply if we replace the individual series- or parallel-connected transistors by blocks of transistors such as the groups illustrated in (c) and (d) which respectively form the AND-OR-Invert (AOI) function, and the OR-AND-Invert (OAI) expression.

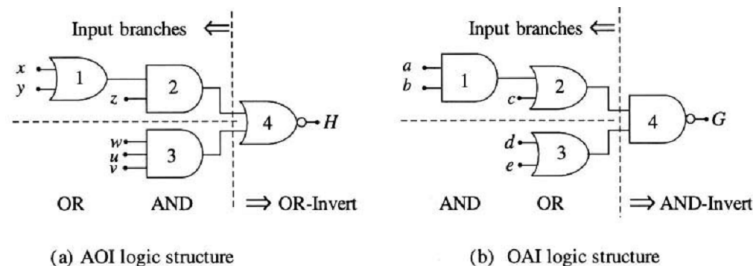


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CMOS COMPLEX LOGIC GATES

Logic Design Techniques

- We can use logic design techniques to formulate or structure the complex logic gates.
- The AOI network (a) has been structured to show the logic patterning OR-AND-OR-Invert, while the OAI network (b) implements a flow of AND-OR-AND-Invert.
- The Input Branches in both drawings consist of the gates that have logic variables applied as inputs. These provide the basis for creating the CMOS logic circuit.

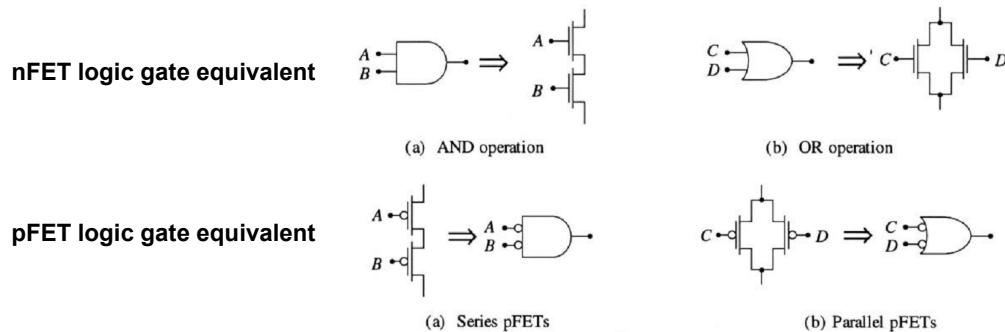


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CMOS COMPLEX LOGIC GATES

Logic Design Techniques

- In general, series-connected nFETs yield the AND operation among the inputs, while parallel-connected nFETs are used for ORing the input variables.
- This provides us with a one-to-one correspondence between the logic gate diagram.
- To give logic gate equivalent or transform equivalent logic gates using De Morgan and Boolean algebra.



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CMOS COMPLEX LOGIC GATES

FET Sizing and Transient Design

- Series-parallel logic gates have the characteristic that the output logic function is determined entirely by the circuit topology.
- The sizes of the transistors do determine the values of the DC switching voltages such as for the various input combinations. However, most of the design and **sizing problems center around the transient switching times** because the use of series-connected FETs introduce delays that are intrinsic to the design style. When analyzing the output transients, we note that
 - The pFET logic array sets the value of t_{LH} and,
 - The nFET logic array sets the value of t_{HL} .
- The procedure for estimating the switching times for a static logic gate can be summarized as follows.
 - Estimate the total output capacitance C_{out} for the gate as the sum of all capacitors connected to the output node.
 - To calculate the charge time t_{LH} , find the longest chain of series-connected pFETs between the power supply and the output node. Construct the equivalent RC ladder network, find the time constant τ_p using the Elmore formula, and then estimate $t_{LH} = 2.2\gamma \tau_p$.
 - Use the same procedure to find t_{HL} . Identify the longest chain of series-connected nFETs between the output and ground, construct the equivalent RC ladder network, apply the Elmore formula to find τ_n and then use $t_{HL} = 2.2\gamma \tau_n$.
 - A computer simulation is required for an accurate analysis.

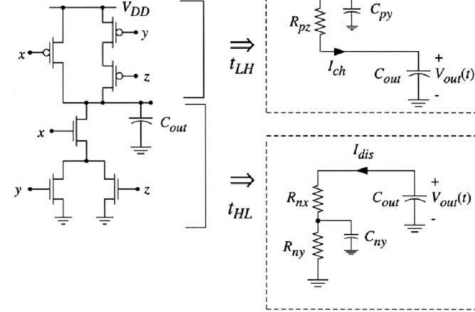
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CMOS COMPLEX LOGIC GATES

FET Sizing and Transient Design

Example: $f(x, y, z) = \overline{x \cdot (y + z)}$

- $t_{LH} = 2.2[(R_{py} + R_{pz})C_{out} + R_{py}C_{py}]$
- $t_{HL} = 2.2[(R_{nx} + R_{nz})C_{out} + R_{ny}C_{ny}]$
- The simplest connection to the device sizes is through the resistance formula:
$$R = \frac{1}{k' \left(\frac{W}{L}\right) (V_{DD} - V_T)}$$



- First, we design an inverter that has the desired values of

$$t_{LH} = s_p \tau_p = s_p R_p C_{out} \quad t_{HL} = s_n \tau_n = s_n R_n C_{out}$$

- The values of s_p and s_n are then used as a basis for calculating the sizes of the transistors in the complex logic gates. For m series-connected pFETs, we choose for individual resistances

$$R_{p,m} = \frac{R_p}{m}$$

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CMOS COMPLEX LOGIC GATES

FET Sizing and Transient Design

Example: $f(x, y, z) = \overline{x \cdot (y + z)}$

- Aspect ratio $(W/L)_{p,m}$ in the series, where $(W/L)_{p,m}$ is the size of the inverter pFET.

$$\left(\frac{W}{L}\right)_{p,m} = m \left(\frac{W}{L}\right)_p$$

- This approach thus equates the end-to-end resistance of the series chain to that of the single pFET used in the inverter.
- The longer chain, the larger factor m , which can result in large device sizes. Similarly, for r series-connected nFETs, the aspect ratios

$$\left(\frac{W}{L}\right)_{n,r} = r \left(\frac{W}{L}\right)_n$$

where $(W/L)_{n,r}$ is the size of the inverter nFET

$$R_{n,r} = \frac{R_n}{r}$$

- The end-to-end resistance of the nFET chain equals R_n .

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CMOS COMPLEX LOGIC GATES

FET Sizing and Transient Design

Example: $f(x, y, z) = \overline{x \cdot (y + z)}$

- Device sizes for pFETs:

$$\left(\frac{W}{L}\right)_{py} = 2 \left(\frac{W}{L}\right)_p = \left(\frac{W}{L}\right)_{pz}$$

- Device sizes for nFETs:

$$\left(\frac{W}{L}\right)_{nx} = \left(\frac{W}{L}\right)_{ny} = \left(\frac{W}{L}\right)_{nz} = 2 \left(\frac{W}{L}\right)_n$$

- Note that the size of the pFET with input x is not specified directly by this procedure. It can be the same size as the inverter pFET, or may be chosen to be the same as the other two pFETs to simplify layout.
- The followings of this design procedure are ignored
 - Some important aspects of the transient design.
 - The inter-transistor capacitance within a series connected chain.
- The most obvious among these is the dependence of capacitor values on the layout dimensions

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CMOS COMPLEX LOGIC GATES

- From Inverter and Complex Logic Gates (NAND, NOR), we can apply similar ways to design other CMOS logic gates.
- From basic to complex logic circuits:
 - Adder
 - Latch
 - SRAM
 - CPU, and so on
- A set of logic gates/cells in digital IC design is called **Digital PDK Library or Standard Cell Library**. It consists of thousands of available logic gates (symbol, layout, netlist) matching the rules for a specific process technology.
- This library is used to synthesis and Place-and-Route in Back-end of Digital IC design process.
- In Synopsys, it is called Milkyway library. Meanwhile, in Cadence, it is called OpenAccess library.

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**THE END
(THIS PART)**

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